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Dimitrij Tschodu

How to Study Physics Effectively and Sustainably

Tips and Tricks for First-Year Students

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Tips and Tricks for First-Year Students

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What You Can Find in This *essential*

- A detailed and carefully worked out experiment protocol of the biggest experiment in the study of physics: the study itself
- An honest description of how to study physics without nervous overload
- An opinion—carefully placed between the lines—that diligence trumps a talent for physics or mathematics

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Without my wife, Anastasia Wolschewski, the key of this *essential* would be neither cheerful nor bright, but rather dull and gloomy.

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Three Hours of Reading Instead of Five years of Stress

1

Summary

A consequence of the many-worlds interpretation of quantum mechanics could be that in another universe a copy of you has just realized a chain of happy life decisions and—receives the Nobel Prize in Physics. In another universe, on the other hand, studying physics could be described as foolproof.

But we are apparently in a universe in which about every third physics student in Germany drops out of his studies [15]. This does not have to be the case.

This thin book contains all sorts of experiences and methods, laboriously and painfully collected, carefully selected and finely honed during my physics studies. Its scope ranges from the usual mistakes how to approach the exercise problems, from tips for group work to detailed descriptions of exam preparation: all this to help you successfully complete your degree.

Learning physics does not have to be incredibly difficult. It really doesn't. You just have to ask the right questions and turn them into action: What mistakes should I avoid right from the start? What am I wasting my time with? How do I approach the exercise problems? How do I learn most effectively for the exam? And finally: How do I learn not only effectively but also sustainably?

I asked myself such questions far too late. So I had to learn from my own mistakes and work hard. A Chinese proverb says that *a wise man learns from his own mistakes, but an even wiser man learns from others*. I want you to be wiser than me and learn from my mistakes.

How to Avoid Mistakes in the Beginning

2

Summary



Many believe that in physics studies, intelligence alone is what counts. But I believe that studying physics is like studying sports: You can be incredibly fast or strong or whatever. But in the end, it is all about technique. In this chapter, we correct the mistakes in the technique of studying.

2.1 Willing to be Intelligent

At the beginning of my studies, I admired my trainers and professors: “Wow! You’re so intelligent!” So I tried to be intelligent too: Every single thing in the lecture had to be understood and researched. That worked—for 1 week. But the lecture material became more challenging and the speed at which the material was covered increased. So something developed in me that psychologists consider a typical example of cognitive dissonance: One realizes that a thing begun becomes more tedious than expected.

But I was not the only one in the club of mental discomfort: truly intelligent students—and not wannabes like me—had a hard time, too. In school, they got used to getting good grades with relatively little effort. Well, that was the end of it. Many of them could not cope with it and dropped out of their studies. This had nothing to do with their intelligence, but only with their willing to be intelligent.

If you are an intelligent person and are constantly trying to live up to high expectations of your wisdom, then stop it. Instead, concentrate on simply not being stupid. This will make your study life easier and more relaxed.

2.2 Willing to be Perfect

There is no place in physics studies for perfectionists. They can get the place, of course, but they will have to wait a very long time. Perfectionism is a good thing in itself: you try not to make mistakes, meet deadlines, and just be reliable and better. But especially the first sign of perfectionism can quickly become fatal: not willing to make mistakes.

I do not know where this idea comes from that mistakes mean something really bad. Maybe it comes from our school days: In school, mistakes were punished by giving bad grades. But those days are over. Now it is all about learning. And the quickest way to learn is to make mistakes and become aware of them. There is nothing wrong or embarrassing about that.

Not to make mistakes is even dangerous. Why is that? You go to every lecture and try to understand every proof. Then you spend the whole day trying to solve an exercise correctly and write it down nicely. In the middle of the semester, you realize that it cannot go on like this: “I’m too slow.” Then you cancel some modules, worry about it, put a lot of psychological pressure on yourself and—think about giving up.

Before the time comes: **Dare to make mistakes—but learn from them quickly.** That way, you will enjoy your physics studies and take away the unnecessary stress.

2.3 Wasting Time on Trivial Problems

You can waste a lot of time with trivial problems in physics degree: A program has to be installed for the internship. Greek letters cannot be displayed in the plot. A course cannot be registered online. Or the best thing: You spend half the day evaluating the results of your practical training and do not save the evaluation: Your notebook suddenly crashes and you want to cry.

So do not waste your time on trivial problems from the very beginning. Find a person who can help you. Someone who can show you how to install the damn program, for example. Even the Terminator got help:

I am not a self-made man. Every time I give a speech at a business conference, or speak to college students, or do a Reddit AMA, someone says it. “Governor/Governator/Arnold/Arnie/Schwarzie/Schnitzel (depending on where I am), as a self-made man, what’s your blueprint for success?” They’re always shocked when I thank them for the compliment but say, “I am not a self-made man. I got a lot of help.”
[12]

It is so easy. Still, it will save you a lot of time and trouble.

2.4 Competing with Fellow Students

Anyone who justifies blind ambition with hackneyed phrases like “being nice gets you nowhere” should definitely not study physics. So far I have only met two such physics students and can assure you: Nobody wants to work with them. With their competitive spirit, they only harm themselves. And that is the beauty of studying physics: You help each other. **So instead of elbowing your way, you should ask yourself how you can help others.** The help does not have to be expressed in the usual group work: Maybe you can do masterful research or have a special flair for the written word. Share this with your fellow students. Think long term.

2.5 Isolate Yourself from Others

This tip is for loners and those who think they are outsiders: Do not isolate yourself from your fellow students. To be honest, especially in the first semester, there are many people who get on your nerves. However, this should definitely not be the reason to isolate yourself and go through the whole study on your own. By communicating with your fellow students, you will also learn important things like old exams or exam-related topics. **Talk to others** and you will see how much more pleasant studying physics can be.

2.6 Read More than Necessary

For certain students, myself included, this is a real problem: There are so many books you could read! First of all, there is the recommended literature, then all the books you use to prepare and follow up the lecture. In addition, the textbooks from which one copies the solutions are always available. And because the calculations are explained there step-by-step, you will want to read these books as well (see Chap. 12).

The curiosity is praiseworthy, but the urge is counterproductive. And even if you do not want to pick up a book at all, you should read the relevant chapters and possibly summarize them. **You also do not have to read the books from the first to the last page.** If your time is important to you, get rid of this habit.

2.7 Being too Reserved

Modest is not the same as reserved. For example, you can be modest but still not reserved. Why is this a mistake in physics studies? Because trainers and professors are people, not machines, who do not make mistakes: whether in correcting your exam or in the exercises.

So learn to be difficult: **Ask questions during and/or after the lecture.** Go to the exam review and annoy the lecturers there with your questions. Stay persistent but respectful: this way, you leave a positive impression for possible projects.

2.8 Not Leaving Time for Experimentation

Every fifth day, Google employees can take care of their own projects: In this so-called “20% time,” the ideas of Gmail, Google Maps and AdSense were born. In physics, however, it is difficult to take time off every Friday to work on your own project. But we can get inspired by Google and find this time elsewhere: During the semester break, for example. So in every semester break, I tried things that motivated me for the next semester: I looked at classic physics problems like the Brachistochrone curve and the Feynman tricks. Or I taught myself the programming language Python to program interesting physical processes. All this was incredible fun and gave me energy for the next semester. **So do not let the norms and obligations of a physics degree suppress you: Experiment.**

Most physics students simply start studying: they do not worry about how they learn best and what mistakes they can avoid. About one third of the beginners break off their physics studies and the rest struggle [15]. Do not make the same mistakes: work on your technique.

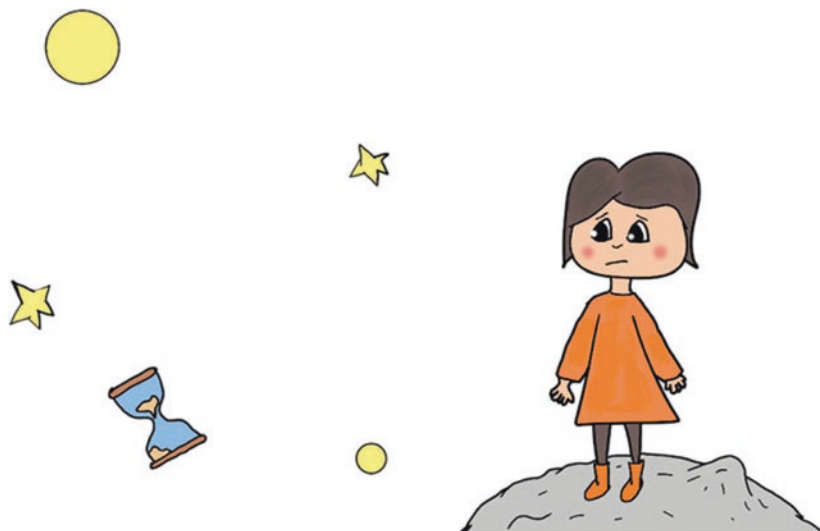
TODO

- Concentrate on not being stupid.
- Dare to make mistakes and learn from them quickly.
- Do not waste time on trivial problems.
- Help your fellow students instead of competing with them.
- Do not isolate yourself.
- Do not read more than necessary.
- Learn to be difficult.
- Leave time for experimentation in your studies.

How not to Waste Your Time

3

Summary



You learn many exciting things during your physics degree: Starting with simple linear systems of equations up to the cool sounding annihilation operators. But I have always been surprised that nobody applies the ideas learned to his or her life. How so? It is simple. For example, you learn to analyze a problem, break it down into its elementary units and find connections. In the first year, almost all physics students have the same big problem: the lack of time.

But instead of following today's madness of time optimization, where you want to get as much and as fast as possible, you should reverse it: **Make sure you do not waste your time instead of optimizing it.**

So in this chapter, you will learn what the most common time eaters in physics studies are and how to avoid them. But before that we should ask ourselves how much time physics studies take up.

3.1 Time Spent in Physics Degree

So how much time is needed? Is that an unbelievable 70 h per week, so that you can forget the three big F's—family, friends and free time?

The amount of time spent on the official events such as the lecture, the exercise or the practical training naturally depends on the individual course. But in physics, you approach everything you can: In the first semesters, there are usually three to four courses with each course consisting of two lectures and a seminar. For one course, 3 times 1.5 equals 4.5 h. If we calculate with four courses, we get about 20 h per week.

And now self-study comes into play: If you are fast and only need 10 h to rework and solve exercises, then, "Bravo!" – you have done it and can hope to win the Nobel Prize. However, most of us—if we really want to learn sustainably and not just copy off—will have to bear the heavy burden of the bell curve with its 30 h. **So all in all, 50 h of work per week await you in your physics degree.**

3.2 Self-Study

The biggest mistake you can make in studying is to rely on it. That is why you should learn most things on your own. That is why you can waste most of your time only in self-study. Before this happens and you start wondering why you are so slow in learning, you should make a habit of the following: Divide your time into blocks.

This may not sound so exciting, but it is extremely effective. Let us take an example: Many people spread their shopping over the whole week and go shopping three to six times. If it takes 10 min to get to the mall, that is at least 1 h to get there and back each week. Even if we neglect the duration of the shopping, this is three times as much as if you only go shopping once a week.

The example does not convince you? Okay, the real reason is hidden in a special feature of studying physics: Here you learn to solve problems by first getting

thoroughly familiar with the theory and then delving into the problems. And this immersion in theory and problems does not require units of time in minutes, but in whole hours and half days. If you are interrupted in this time span, even for a few minutes, the Ariadne's thread slips away—you get lost in the mathematical labyrinth of the task and have to start all over again. Time is lost.

But let us get down to business: How do you divide the time of day into blocks? How do you proceed and what is your guidance? **First of all, you look in your timetable for large open windows of time in which you can work without any distractions:** For example, if your first lecture does not start until 11 a.m., then you could study intensively between 8 a.m. and 11 a.m.

Second, you should stack your problems according to the topic. For example, it would be more effective to do the reworking of the math lecture and the solving of the math problems in one time block, because here the temporal and intellectual gain is greater than the sum of its individual parts—this is called emergence.

Third, the time blocks must be prioritized and distributed throughout the day. What does this mean exactly? This is where your self-perception is needed: When are you most concentrated and when are you totally exhausted? You could, for example, read numerous textbooks and ask experts when is the optimal time for cognitive performance and so on. But that takes time. So why not act like a physicist: Take a laboratory book, that means a normal pad, and write down for the next 2 weeks at what times you can work most concentrated, full of energy or in a good mood, etc. Knowing this, you can now divide the time blocks according to your analysis: If you are full of power in the morning and have the greatest difficulty with math, then the morning is exactly the right time for it. In the late afternoon—hovering between intellectual fatigue and nervous listlessness—you could, for example, do light experimental physics exercises. (With all due respect to experimental physics: In my opinion, an easy problem in theoretical physics is similar to a difficult task in experimental physics).

Finally, never underestimate the power of pessimism. In other words, you should not have too many items on your checklist for the day: Not dumb is better than intelligent, and less is more. **Concentrate on a few tasks such as *reworking math and solving math problems* instead of writing down many things you cannot do and will be annoyed by.** Follow my favorite law, Hofstadter's law: It always takes longer than you think, even if you consider Hofstadter's law [17].

3.3 Television, Social Media and Magazines

Social media like Facebook or Twitter can be very helpful. After all, you can form study groups there to exchange thoughts, problem solutions and other useful resources. You can also follow motivating and helpful websites or people. However, you should use them with caution during the learning period. Because the biggest danger in studying is procrastination, which has become fashionable. However, it is the job of Facebook and others that you visit their websites and thus procrastinate.

3.4 Group Work

Group work (or example tutor or seminar groups) can give you the most memorable and fun moments of studying physics: You remember with pleasure and a certain pride long afternoons filled with suffering and headaches. But this memory is only pleasant when you have successfully completed your studies. However, group work can be one reason why you stop studying physics.

Do not let students and professors intimidate you by saying that you should definitely solve the weekly exercises in a group. This statement is not always true. You just do not know from the outset how useful group work will be. That is why you should test it for a few weeks, paying attention to the following three “if” statements:

- **If you feel stupid about arrogant fellow students.** It is perfectly okay to have that feeling in the lecture or seminar. But the purpose of the group work is not only to solve the problems: It is rather the feeling that you are not the only one who cluelessly and blindly copies off the blackboard. However, if you feel exactly the opposite of this feeling, you should ask yourself where it comes from: Is it the possible arrogance of your fellow students or just your insecurities? If it comes from arrogance, you should leave this group as soon as possible: There will be plenty of times in physics class to feel stupid.
- **If the tasks are solved by a few smart students who do not bother to explain it to you.** This often happens: one or two really bright minds enlighten the whole group. The only problem is that, as in every cult, the words of the enlightened ones are followed blindly. At the end of the semester, you fall through the exam and wonder why, because actually, you were solving the

hard problems together. But you were not. So make sure that you really calculate and understand the exercises in the group and do not just copy them.

- **If the group work is poorly organized.** Good organization means that all participants must be reliable: No one is late, just copying notes or acting important. If any of this is not true, then the group work is poorly organized. You can of course try to organize the group work yourself to avoid the mistakes mentioned above. Ideally, the duration of the work should then be fixed in advance to avoid digressions during the group work: After all, students tend to lose sight of the physical problems in order to have a chat. However, if there is too much chatter, you could intervene in the chat and say the following, for example: “I am sorry to interrupt you, but can we get back to the problem? Because I have to leave soon and it would be nice if we could do it today.” A psychological study has shown that people are more likely to do something if you give them a reason—the reason itself is completely irrelevant [22].

Whether it is optimizing time or avoiding wasting it—in the end, it is about spending more time with what we love: physics.

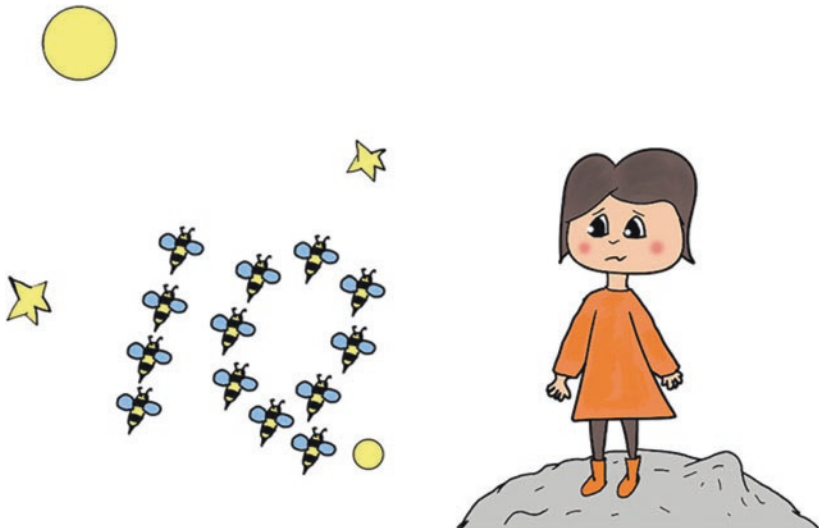
TODO

- Find large open time slots.
 - Prioritize them throughout the day.
 - Stack your problems according to the topic.
 - Concentrate on a few tasks.
- Be careful with social media: danger of procrastination.
- Find out if group work is right for you: Test it.

How to Make Your Studies Easier

4

Summary



There are things you know and things you do not know. To know means to have knowledge of something. Since we know almost nothing at the beginning of our studies, we do not know anything either: We do not know how to solve problems, which books to read, how to prepare for the labs, which information is important—there are many more things we do not know. Therefore, one starts to look for the right way in the jungle of not-knowing.

There is now the possibility to simply repeat what most others do: Read the same books, visit the same study group, give the same solutions and—make the same mistakes.

Then there is the possibility: Isolate yourself and learn on your own to get what you think you deserve: Brilliant grades. The danger: You neither help your fellow students nor yourself, because through isolation you miss useful information.

The third possibility: learning from and with others. But also to study independently and effectively by learning to distinguish useful from useless information, to research quickly and purposefully and to use the power of swarm intelligence. This is exactly what this chapter is about.

4.1 Surround Yourself with Fellow Students Who are Better Than You

Investor legend Warren Buffet silences Wall Street when he has something to announce. But one of his wisdom is also helpful for physics students: “It’s better to hang out with people better than you. Pick out associates whose behavior is better than yours and you’ll drift in that direction” [2].

For us, this means: Surround yourself with the best students from your course. If they are the best, they do or know something that you are missing. Maybe they read special books? Or they watch certain online lectures? Maybe they are taking a tutorial you have never heard of? Ask them and find out what makes them so good. If that does not work, work with them—unless they make you feel stupid—and try to understand how they approach physics problems.

4.2 Research Like an Investigative Journalist

Fellow students who are better than you are also your tutors—they just live in a frame of reference that was shifted a few years during their studies. Therefore, they are your most valuable informants. Ask them the same questions as well as the best students in your year of study. However, you should not ask professors these questions. Why? Because their years of study are far in the past—they may have forgotten what they struggled with as students. But good answers to these questions require sensitivity to the suffering of the young physics student.

Besides the best students and tutors, there is another great source: the student committee. The student committee is great because it archives old exams and has useful insider information.

In the rare case that there are no good students in your year and both your tutors and the student committee are silent as the grave, as if they were being interrogated by evil professors in the next room, then you should do something that we always do when we do not want to do: google it. But google cleverly and use the so-called Google search operators. For example, you can use the English quotation marks to find results that match the exact phrase in the search entry. Why is this useful? Professors are people and people are energy-conserving animals: The same physics problems circulate in universities and on the Internet—someone has thought them over and others have copied them. So if you look for the exact word choice in one of your exercises, you will find similar websites of other professors. There you might find solutions and relevant material for the exam.

There are of course many other useful search operators: just google them!

4.3 Use Swarm Intelligence

Swarm intelligence here means that you find the best material with the help of other people. This could be a book, an online lecture or a lecture script.

Suppose I am in my first semester and I want to find an excellent book on the Fourier transform, which I need for the preparation of my practical course. I have already read the description of the experiment and did not understand much. Here follows the procedure:

Google is my best fellow student: I type “Best Book Fourier Transformation” into the search engine and write down the first five hits:

1. *Fourier transformation for engineering and natural sciences* by Bruno Klöng
2. *FFT applications (FFT-Anwendungen)* from Elbert Oran Brigham
3. *Fourier analysis and distributions (Fourier-Analyse und Distributionen)* by Rolf Brigola
4. *Laplace and Fourier transformations* by Otto Föllinger
5. *Laplace, Fourier and z-transformation* by Hubert Weber and Helmut Ulrich

The next step is to look for a recommendation in the Internet forums: We google “Best Book Fourier Transformation Forum.” Click-Scroll we get the following recommended:

6. *Fourier transform for pedestrians* (*Fouriertransformation für Fußgänger*) by Tilman Butz is mentioned in two forums. One of them with the comment that it is not suitable for beginners.
7. *FFT applications* by Elbert Oran Brigham are mentioned and commented in a forum: There is a detailed explanation of the algorithm and a pseudo-algorithm—very good if you have to program for the practical course!
8. Someone writes that a book is “a total overkill”—unnecessary for the subject. Other forum members also recommend only short resources like the Wikipedia article on the Fourier transform. This should be kept in mind.

Last but not least I search the social news aggregator *reddit.com*. The answers here are rated by members, so the best answer is determined collectively.

9. The best answer is that one should not look for a specific book, but rather in the relevant chapters of standard physics textbooks. The second-best rated answer says the same and recommends *Mathematical Methods in the Physical Sciences* by Mary L. Boas.

What do I do now? I note the books in 1, 2 = 7, 6, and 8, go to the library and look at them. I skimmed the texts and came to the conclusion that I find *Fourier transformations for pedestrians* very practical and that I should read it sometime. For the practical course, the corresponding chapter in *Mathematical Methods in the Physical Sciences* is suitable and indeed sufficient.

Ten minutes of good research and a visit to the library saved me about a day of hard preparation and occasional verbal abuse of the person who wrote my lab manual.

At the beginning of the chapter, I said that there are rather things you do not know—and therefore do not understand—when you start your degree. That is why during the whole study, you believe and hope to experience the moment of sudden revelation, after which suddenly everything makes sense and all connections are clearly visible like the stars on a cold winter night. This moment can come. Or not. But one should work toward it—you will find more of this in Chap. 9.

TODO

- Find the best material for yourself.
- Surround yourself with smarter fellow students.
- Learn to research effectively.
- Use swarm intelligence in your research.

How You Concentrate Intensely

5

Summary



A student comes to the library at 9:00 a.m. He unpacks his stuff. Before preparing his desk, he looks at his cell phone and types in a message. He puts his cell phone aside and starts working. His smartphone vibrates. He must be answering. It is 9:30 a.m. in the morning. But he forgot to check his e-mail. Now he is writing an e-mail. It is 10:00 a.m. Get back to work. At 10:31 a.m., his stomach growls. He leaves the workplace. An hour goes by and he comes back. Full attention. The

library slowly fills up and a pretty student sits right across from him. He looks at her periodically. It is 12:30 p.m. At 1:15 p.m., he has to attend a lecture. There is no point in going on working: He is surfing on the Internet. At 1:00 p.m., he leaves the library.

Formally, the student was busy for 4 h. But in fact he only studied for 31 min. Moral of the story: I should stop being distracted by such students because I myself did not even work 31 min with concentration while watching him.

This chapter is therefore about a secret superpower: being able to concentrate intensely. Why is this a superpower? Because it allows you to reduce the whole day of study to a few hours. Calvin Newport—professor of computer science and author—puts this into a useful formula [27]:

$$\text{High-quality work performed} = (\text{time spent}) \times (\text{intensity of concentration}).$$

The more intensive the concentration, the less time we need for learning if the work is kept constant. As nice and simple as this formula may sound, experience shows that a very difficult task must first be overcome: sit down and start learning. That is why I mention three “variables” on which the intensity of concentration depends. They will help you to trick yourself and get the job done.

5.1 Time

In Chap. 3, we talked about time blocks. **The most important thing about these time intervals is 100% concentration:** No emails, no smartphone, no Facebook, etc. Often it is very difficult to be distracted by nothing like a Jedi Master. For example, I would do anything not to get my work done: Get another cup of coffee, think about life, or stare at the rough white blanket over my head. That is why you should design your environment in such a way that you avoid your triggering sensations: For example, install an app or an extension that for a certain amount of time blocks all websites that you should avoid during the learning interval. Or warn your roommates that you are physically present but mentally absent at certain times. You can also change your environment completely: Go to a place where you are not distracted.

5.2 Location

Warning: A crowded library is not a good place to learn. Designed for learning, it is hardly possible to learn there. Because we are social beings and therefore pay special attention to our social environment—at least in my experience. Also, in almost every crowded library, there are students who yawn loudly, rustle with a pile of paper, or hiss behind your back, but if you are a better learner with background noise, the library and cafes like Starbucks are the perfect place to learn.

Otherwise, it is worth exploring fairly empty branch libraries with little-known workstations. **Whether it is an unfamiliar branch library or a highly desirable learning booth, there is a danger of feeling busy in such environments without really learning effectively—you should be aware of this danger.** Ultimately, the choice of your learning location depends on something very important: your learning routine.

5.3 Routine

Famous people differ in their working routines: some writers get drunk or massage themselves at certain erogenous zones to get into the writing process [8]. What all great personalities have in common, however, is the constant repetition: it is absolutely unimportant what your routine looks like—the main point is that you carry it out consistently. In the following I explain why it is so important for physics students to have a working routine.

The phenomenon is called *regression to the mean*: Consider a measurement. If a first value is extreme, then the following value will be closer to the average, if the measured values are influenced by chance. This has nothing to do with causality—the regression is purely random [33]. For example, if your motivation is at its peak for a while, it is likely to become mediocre again. Conversely, if you experience a low point in your study life and decide to increase your motivation with energy drinks, you might become more motivated. The question remains as to how much the energy drinks have actually influenced the increase—it is more likely that they have nothing to do with it.¹

¹Example inspired by Daniel Kahneman's example on page 183 in [19].

In short, your motivation will come and go. Therefore, you should not hit the bad days with frustration and doing nothing, but with autopilot: i.e., with work routines that make learning easier while your motivation is absent.

In physics, it is particularly easy to introduce persistent routines. After all, every semester we have exercises that are given out on the same day of the week and at the same time. Let us imagine, for example, that the weekly exercise series is handed out on Monday at 9:00 a.m. and you have a free period every Monday between 4:00 p.m. and 8:00 p.m. Every Monday, just before 4:00 p.m., you make yourself a coffee, drink it, listen to one of your favorite songs, and go to the library. There you solve the exercise problems until 8:00 p.m. That is your routine on Monday.

What is so great about it is that it resembles the classical conditioning according to Pavlov. Ivan Pavlov is the man with the dog and the bell: whenever the bell rings, the dog gets the food and is happy about it. Next, the bell rings, but the dog does not get any more food. Nevertheless it is happy, i.e., it lets his saliva flow [30]. Pavlov has tricked the dog. In a similar way, you can trick yourself. In your experiment, you are both the dog and the physiologist. The smell of coffee, the positive association with the song and the library are the stimuli and working on the exercises is your reaction: even if you do not feel like doing the exercises, it might be easier for you to start working because you are used to it.

The transfer of classical conditioning to humans is often criticized [20]. But there is also the view that it is mainly applicable to the human learning process [36]. My advice is to try it for a few months and see if it works for you because *all life is an experiment. The more experiments you make the better* (Ralph Waldo Emerson).

5.4 Practical Elements of the Routine

It depends on your preferences how, where and when you learn. However, before you start the exercise problems please note the following four things.

First, you should usually start with the more difficult problems, because your concentration will drop. Unless you calculate the easier problems in order to get used to them. But the familiarization should not take long. Furthermore, difficult tasks are given more points than small arithmetic bites, which can sometimes be very annoying because you get the easy solutions very slowly.

Second, you should fully concentrate on the process. Why is that so important? Anyone who has taught him- or herself a little speed reading knows this simple trick: Run your finger over each line. This avoids the eyes jumping back

and forth, thus increasing the reading speed without affecting comprehension [4]. It works similarly with thoughts: The focus is on the solution process and not on the solution itself. Do not think about the solution during the solving process and thus avoid the thoughts jumping back and forth.

Third, it is better to calculate the problems as soon as possible after releasing. Why? Because in the remaining time, you can not only compare your solutions with those of your fellow students, but also discuss them. This is difficult, but it helps enormously to understand the actual physics behind the exercises and to keep solutions in mind for a long time: Hard semester—easy exam, easy semester—hard exam.

Finally, the so-called Pomodoro technique is a useful learning tool in the study of physics. This technique is a time management method developed by Francesco Cirillo, who invented it in the 1980s when he used a kitchen clock in the shape of a tomato to work—hence the name (Italian: *pomodoro* = tomato) [6]. The Pomodoro technique is usually broken down into four to six steps: First you define your task. Then you set the stopwatch to 25 min. In the third step, you work on your task. In the next step, you take a 5-min break. After four such 25-min work phases, you take a longer break, which can last 15–20 min [6].

However, the technique for solving physics problems should be changed slightly. So you do not need the first step to solve exercises, because the exercises are usually clearly formulated—you just have to solve them. In addition, experience shows that 25-min intervals are too short. Especially for the problems in theoretical physics, where you sometimes need half an hour to understand the task, 25-min intervals are too short. But they are perfect for solving easy tasks in experimental physics. One could of course object that the effectiveness of the Pomodoro technique lies in breaking the task down into smaller subtasks. However, a physics student often does not know which subproblems will make up a difficult physics problem: Every now and then, for example, you will have to solve complex integrals. And to do so, you will first have to rewrite them in polar coordinates. To avoid making mistakes, you will have to google them again. After you have solved the integral in polar coordinates, you will have to look up the solution of the Gaussian integral. This procedure can go on for a long time.

For this reason, a longer Pomodoro phase is more useful for you. I find a 45-min work phase with a quarter-hour break particularly effective. However, the Pomodoro technique does not always prove to be efficient. For example, you do not need to take a break if you are in the middle of solving a task and are eager to find the solution. On the other hand, this technique is enormously useful for exam preparation. Because in this case, you know the extent of your tasks and how they

are structured, since you have already solved them or similar tasks—at least you should have done so during the semester.

If determining the time, place and a routine including the Pomodoro technique sounds too complicated to you, remember the above story with the student and try to keep the formula mentioned at the beginning of the chapter in mind:

$$\textit{High-quality work performed} = (\textit{time spent}) \times (\textit{intensity of concentration}).$$

It will give you a good guidance throughout your self-study.

TODO

- Perform quality work.
- Do not be distracted: 100% concentration is required.
- Design your environment accordingly.
- Introduce routines to stabilize your learning phases.
- Start with the more difficult tasks.
- Focus on the process, not the result.
- Calculate the tasks as soon as possible after their output.
- Try the Pomodoro technique.

How to Understand the Math Lectures Faster and Better

6

Summary



It is the best and the worst time, a lecture of wisdom and incomprehension, a language of sentences and proofs, a semester of hope and a test of despair¹: it is

¹This sentence is a homage to the first movement in *A Tale of Two Cities* (*Eine Geschichte aus zwei Städten*) by Charles Dickens.

mathematics that makes the first year so dramatic. But before you feel discouraged or despair of the lecture, let me reassure you:

- Nobody understands the lecture 100%.
- Hardly anyone understands most of it.
- For most people, only some parts of the lecture are clear.
- Hardly anyone can think of the solutions immediately.
- Everyone has difficulties with the exercises and only a few come up with the right solutions.
- It takes a lot of time and effort to understand proofs and derivations. But you will be able to calculate practical things like differentials, integrals, operators, etc.
- Not understanding is a necessary criterion for comprehension.
- At the end of your studies, you will not have understood everything. However, you will have the feeling that you can understand it if you just wanted to get used to it.

But we are not yet at the end of the study: We sit in the mathematics class and try to decipher the salad of Latin and Roman numerals, ancient Greek letters and extra-terrestrial symbols on the blackboard. Here are now the rules to make the lecture faster and better understood.

6.1 Keep Good and Detailed Lecture Notes

Sit in the first row, stick your tongue in the left corner of your mouth and write down the lecture as if your life depends on it. You will be very grateful to yourself later: Because when you are rewriting, it is easy to stumble upon little mistakes that can happen if you are not careful when you copy from the blackboard. For example, a wrong accidental seems to be a trifle—until the transcript does not make sense anymore and you have spent half an hour in the maze of accidentals. You should also pay attention to the following points when taking notes:

Manual labor is sustainable

Avoid typing the lecture into the laptop: First of all, it is extremely time-consuming—no matter how fast you can hit the keys, because formulas and graphics cannot be typed in very quickly. Second, several studies have shown that handwriting has positive effects on learning and long-term memory [7, 26]: Handwriting is sustainable.

Encryption is secure

Also, introduce your own symbols that encode your script at the edge of the page: a lightning bolt, for example, could symbolize a contradiction that you should definitely resolve. A magnifying glass could represent an imperative to look up—something you should google, for example. Which and how many symbols you scribble on the edge of the page depend on your creativity and way of thinking.

But there is one symbol that you should never miss: the question mark.

Of course, this refers to something you do not yet understand. Studying physics can be incredibly fun if you think of it as a mission where all the question marks that accumulate during this time have to be answered: For example, as a still unknown student, the fantastic Mr. Feynman took a notebook and wrote down things he did not understand [14]. The notebook had the title *Notebook Of Things I Don't Know About*. Use your encryption to make sure that you know what you do not know.

Aggression is necessary

You should not act aggressively toward your fellow students or professors, of course. It is about you being aggressive with your script. What does that mean exactly? Your script should not be an untouchable sacred scripture from which golden rays radiate, illuminating you in a flash. **Rather, you should actively take notes: comment, formulate associations, and carefully control what you write.** Do not think of the transcript as a piece of paper, but as an opponent in the conversation about the truth.

However, you should be careful about marking and underlining important sentences or statements: it may be that attention is directed to individual elements rather than to the connection between them, thus only giving the impression that you have understood something [10].

Always keep your eye on the big picture

Most mathematics lectures have the same logical structure: definitions are followed by propositions and proofs. Then the calculation is done next to it—i.e., secondary calculation is introduced—and examples are pre-calculated, which are rarely useful for the exercises because of their simplicity.

At the same time, one is often up to one's neck in the forest of numbers and figures and no longer knows what the lecture is actually about. **To avoid this happening so often, you should use the time when the professor is writing down the next secondary calculation or presenting a proof to identify the big picture:** What are we trying to prove? Where did we start? How does the proof relate to the starting point? From the main topics the rest branches off, which is at first secondary for understanding and on which you should therefore concentrate later.

6.2 Thoroughly Rework the Lecture

Especially when reworking, you should not let your concentration be diverted to other interesting topics in the textbooks. Therefore, guide yourself on your script: It is the basis for your exam. When editing, you should pay attention to the following points:

It is a good idea to ask why, like a child [10]: Why do I prove it? Why must this condition be fulfilled? Furthermore, you could rephrase whiny statements like *I don't understand this!* into elaborate questions like *Why don't I understand this? Elaborative*, because this method is called *elaborative interrogation* [10].

Second, don not get stuck: Google if you do not know something for sure. For example, you can stare frozen at your lecture notes for several minutes or try to get them to talk by giving them a scary look. But you are wasting your time. Instead, you should look up the hyperbolic sine wave immediately if you do not know how it behaves in infinity.

Third, you could try the following two methods when reworking:

The Panda method	The point of this method is that you try to guide yourself as much as possible by your script and as little as possible by secondary literature. You edit it very thoroughly so that every step and every word must be understood: If you do not understand something, you must look it up immediately. Only then may you return to the script A mnemonic help to remember this method is to call it the Panda method (Both terms—the Panda and the Roe method—were inspired by Andrew Ng, who used them in the context of machine learning [28].): A mother panda chooses a cub and nurtures it for a long time. (Biologists call this the K-strategy.) Similarly, a panda decides to take notes and nurse it for a whole semester
The Roe method	With this method, you are guided by your notes, but you invest enough time in editing many other sources such as books or websites. (Mnemonic help: Many eggs in the roe of the female fish.) By comparing various representations and spellings of the same context, you will crystallize its core. For example, if you wanted to teach a young child what the letter A looks like, you could show him or her the same picture over and over and explain for a long time what the essence of the letter A is supposed to be. Or you could take many contrasting pictures—italic, crooked, colored, small, large, etc.—and the child will guess what the insignificant differences are and what the letter is all about. Similarly, if you are learning from different sources, you will contrast the topics in your script. This method is based on the so-called <i>principle of contrasting cases</i> , which can help students to analyze empirical data [31, 32]

Fourth, you should not just read the textbooks either: Briefly skim the chapter, the text or the calculation and ask analogous questions as in the section *Always keep the big picture in mind*: What are you looking for? What do you want to calculate or prove? What is the result of the whole calculation? Only then do you move on to specific questions and steps.

Finally, talk to yourself. Dina Miyoshi—Professor of Psychology at San Diego Mesa College—recommends that her students to practice self-talk: Initially full of doubts, they later report how well this works [29]. In fact, I have seen for myself how helpful it can be when you get stuck on a task and cannot get anywhere. Thinking aloud will magically draw attention to gaps in logic and careless mistakes: **only conversations with fellow students can be more effective than talking to yourself if you are stuck on a problem.**

Whether panda, roe or other methods: Do not be bored. For the arch-enemy of learning is not laziness or stupidity, but boredom. Vary and experiment with your methods to keep the lecture and the study in general exciting and thrilling.

TODO

- Keep a good and detailed record.
 - Write by hand, not on the laptop.
 - Introduce your own abbreviations and symbols.
 - Write aggressively.
 - Try to keep your eye on the big picture at all times.
- Review the lecture thoroughly.
 - Always ask why.
 - Google, if you do not know something.
 - Use the Panda or the Roe method.
 - Edit textbooks in a planned and practical manner.

How You Crack Problems and Get Better

7

Summary



This chapter deals with the questions of being or not being in physics: How do I approach physics problems and how do I get better? To answer these questions, we could learn something from a billionaire and a philosopher: mental models.

7.1 Mental Models

Ludwig Wittgenstein—one of the most influential philosophers of the twentieth century, who also could whistle masterfully—popularized the concept of so-called *mental models* in philosophy [37]. And Charles Munger—the advisor of billionaire Warren Buffet—considered them the necessary basis for the acquisition of worldly wisdom: “You must learn the models so that they become part of your repertoire” [1].

Even if wisdom on a global scale is not the primary goal of a local physics student at the beginning of his studies, mental models can be very useful for learning new physics concepts [21]. So what are these mental models?

We physicists would call them heuristics, others simply principles, and some worldly wise people would call them homomorphisms between reality and thinking, that is, mental images of a part of reality. They are therefore simplified and reduced explanations of how something works. Whatever they are called, it is important for us to understand how they are formed.

One can acquire a mental model in three steps:

1. **Understand the meaning:** Look at a concept or idea and try to crystallize the core. The following questions can help you do this: Do I understand every word? How is that really defined? Do I need more information? What are the causes and what are the consequences? What do I not understand? Is there an example that refutes my solution? Can I think of the simplest example for this problem? Why should I understand it? Can I express the meaning of the idea in a short sentence?
2. **Simplify and reduce:** Sometimes one decorates physical ideas with complicated-sounding terms, neurotically tinged mathematical rigor and confusing illustrations. But consider an idea as something that needs to be peeled out precisely: Is there something that clouds the clarity of the idea? What are the simplest elements to which I can reduce the concept? To what extent is it relevant to the task, my understanding or even life in general?
3. **Model reality:** Formulate the idea in your own words, combine it with a lively and vivid picture and draw it. The usefulness of the idea should be emphasized.

Especially the last step—the drawing or painting of a vivid picture—is an effective mnemonic help to keep concepts in mind for a long time and to recall them faster from memory [34].

Let us take Newton's second law for clarification: $\vec{F} = m \cdot \vec{a}$.

1. **Understand the meaning:** Obviously the formula says that force equals mass times acceleration. But that cannot be the underlying meaning, because I only say what is written.

The arrows above the acceleration $\vec{a}$ and force $\vec{F}$ mean that these quantities are vectors. A vector is defined as a quantity (number) plus direction. But there are crooked ways: Do crooked vectors also exist or do all vectors have to be straight? After a short look I learn that vectors are actually elements of a very abstract structure called vector space. But the vectors of this space can also be functions or other things. The usual description of vectors as a straight-line displacement of a point in a certain direction only refers to the Euclidean vector space, in which everything is so beautifully illustrated.

It turns out that mass is not equal to weight, because mass is a measure of how much matter is contained in a body: it is therefore the same on another planet. However, we call weight the force of weight, which is about six times greater on the earth than on the moon, for example.

Okay, I understand every word and understand what vectors are. Now this seems simple: If I change the direction and the magnitude of the acceleration, the force will change accordingly—with the proportionality factor m . But what is the cause of the acceleration? Where does it come from?

Acceleration is the change in speed. Where does velocity come from? This question explains Newton's first law, which states that the speed of a body remains constant when no force is applied to it [3]: It remains at rest when it was originally at rest and moves at constant speed when it was originally moving. Consequently, the change in speed is caused by an external force. For understanding, I reformulate the formula: $\vec{a} = \vec{F}/m$.

2. **Simplify and reduce:** I forget for a moment vector space, Newton's first law and the arrows, and use only numbers: If I increase the force at constant mass, the acceleration will also increase. Conversely, a larger mass leads to a smaller acceleration if the force is kept constant.
3. **Model reality:** For example, if I want to jump higher, I should of course train hard. But I should not just start to move a lot of heavy weights. Because the ability of self-acceleration is equal to my strength divided by mass. That is why I should not do bodybuilding, but a strength training where your own strength increases without gaining weight. Alternatively, I could fly to the moon and jump there like a happy moon man. Both: I draw the weight lifting and jumping on the moon as mnemonic help.

In each lecture, mental models are presented: The only thing you should do is to understand them and record them briefly and clearly. Now we know how mental models can be formed and collected throughout the course of study. But how do you apply them to the exercises?

7.2 Cracking the Tasks

The mental models represent the first step in the following formula, which can be helpful for the approach to the tasks:

1. What mental models are involved?
 2. What exactly is required?
 3. Translate normal language into physics language.
 4. Sketch the problem.
 5. Solve the problem while checking the units: A solution is only correct if all units are correct.
 6. Check and improve your solution.
-
1. **What mental models are involved?** Most of the time, the exercises are about the topics that have already been covered in the lecture—at least that is how it should be. If you are present in the lectures and/or work through them thoroughly, identifying mental models should not be a problem: For example, an inclined plane problem indicates that you are dealing with forces—so you should apply Newton’s second law. But there are also cases where you sit desperately and hopelessly in front of a problem and do not know how it relates to the lecture: in this case, look at the given variables and ask yourself in which equations they occur. This could give you an indication of which mental models are involved.
 2. **What exactly is required?** What exactly must be calculated?—asking these questions sounds obvious. But some tasks are formulated in encrypted form. Besides, there is rarely the or a right way to solve the problem: For example, there are several ways to prove something. Therefore, identify what is asked and make it clear how you want to answer it: whether you could equate, differentiate, integrate or transform something.
 3. **Translate normal language into physics language.** Yeah, you read it right: Translate normal language into physics language and not the other way around. This means that every statement in the assignment should be for-

mulated mathematically. If you are talking about maximum speed and two external forces, then write down immediately $\vec{v}_{max}, \vec{F}_1, \vec{F}_2$. Because often the solution of the problem is already in the problem definition: When you write in variables, you notice the hints. For example, the maximum or minimum of a quantity, like speed, implies that a function must be differentiated and set to zero. Therefore, pay attention to such hints.

4. **Sketch the problem.** A sketch says more than a thousand variables. This not only gives you an insight into the physical process of the problem, but you also cultivate a physicist's way of thinking step by step: And the more sketches you make, the faster you get a feel for physics.
5. **Check and improve your solution.** This step is crucial if you want to improve quickly: Check and compare your solution with a correct solution. Normally, all exercises are calculated in seminars. Therefore, you should never skip a seminar if you are not sure of your solution or if you have no other possibility to find a correct solution. Direct feedback is an essential factor when it comes to expertise and performance [11]. Therefore you should not only compare your solution, but really try to improve it: Maybe there is a much easier and more elegant way? Or a general approach?

The approach described is similar to *deliberate practice* [11], which in short is about having a clear goal, getting immediate feedback and learning from mistakes. Learning should be conscious and with the highest concentration. This is exactly what we try to achieve.

7.3 Learning for the Long Term Instead of Passing in the Short Term

The greatest practical potential of the mental models and the described approach is shown at the end of the semester, when the exams are approaching and you suddenly realize that you should have started preparing earlier.

Relatively little effort during the semester will save you quite a lot of stress in the examination phase if you keep a list with three columns on the side: **The first column contains the number or the name of the task, the next one the mental models you need and the last one the tools you use.** An example is shown in Fig. 7.1 The latter are the mathematical tools you need to solve the tasks: cross product, gradient, divergence, rotation, chain rule, spherical coordinates, the Feynman parameter, Gauss's integral theorem, etc. Later on you learn much cooler and science-fiction sounding tools like creation and annihilation operators, Lorentz transformation, density matrix, delta function and renormalization.

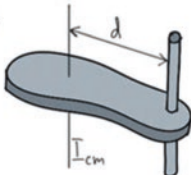
Problem	Mental model	Tools
1	Steiner's theorem $I = I_{cm} + md^2$ 	• $\iiint dx dy dz$ • $\rightarrow \iiint dr d\theta dp$ • Inertia tensor θ: $\theta_{ij} = \hat{e}_i \theta \hat{e}_j$ •  Tensor ! • Cross product, Transformation in polar coordinates
2	Preservation of angular momentum	
⋮	⋮	⋮

Fig. 7.1 Maintain a list with three columns: Task, mental model and the tools used

It is worth destroying the Death Star using the annihilation operator. No, that is not what I wanted to write: It is worth writing down the mental models and the mathematical tools to make sure you master them. At the end of the semester, it turns into a useful checklist for the exam.

Even if you have read the chapter, it will certainly happen that you are still sitting at your problem and will not understand it. In this case, you can rack your brains for a long time, try different approaches and eventually you will solve the puzzle. The question remains how long it will take. Or you can do something outrageous: Write it off. And this is what the next chapter is about.

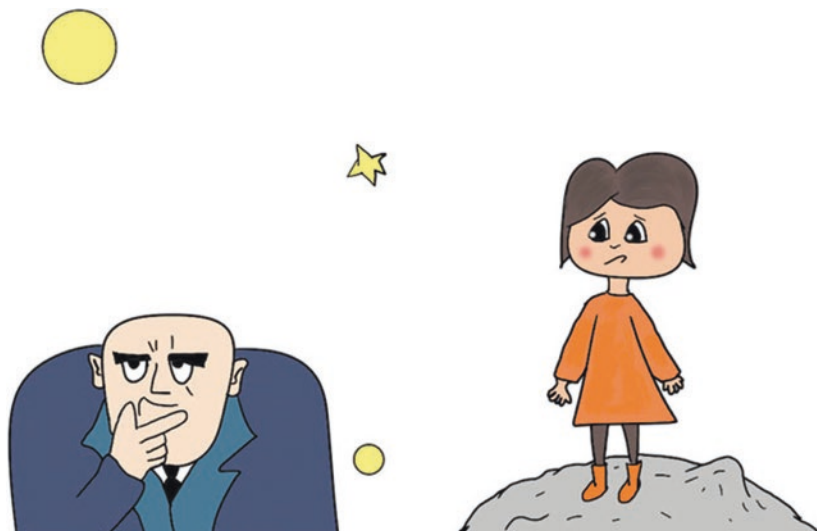
TODO

- Identify the mental models.
 - Understand the meaning.
 - Simplify and reduce.
 - Model reality.
- Understand what is asked exactly.
- Translate normal language into physics language.
- Sketch the problem.
- Check and improve your solution.
- Keep a list of mental models and mathematical tools.

How You Really Cheat

8

Summary



The shameful silence must be broken: Many physics students copy their solutions. Normally, we always hear the same categorical imperative that we should not copy from the sample solutions under any circumstances, and feel the moral finger pointed directly at us.

But this imperative has its dangers. Because if a fresh student tries from the beginning to solve the exercises absolutely independently, she will have to fight

a war on two fronts: With the enormous amount of work on the one hand and the absolute ignorance of how to approach physical problems on the other. The former urges her to complete her problems quickly and the latter advises her to take deceleration measures, to stop and work through them thoroughly.

This tension should not be an excuse for stupidly copying other people's solutions in order to be admitted to the exam—besides, this attitude will fail you anyway. It is rather a matter of relaxing the cognitive dissonance that has been generated. **Students who “cheat” fail in physics studies not because they copy, but because they copy wrongly.**

In this chapter, I explain how to copy correctly.

8.1 Even Benjamin Franklin Did that

To solve the problem of correct writing, we can learn something from Benjamin Franklin—perhaps one of the last polymaths.

His autobiography is one of the most read autobiographies in history [18]. Although stylistically criticized by some, his writing style is simple, clear, pragmatic and even journalistic [18]. How did he acquire this style?

The answer to this question is provided by his autobiography: to shape, grind and polish his style, he read the *Spectator*—a daily newspaper produced between 1711 and 1712. Franklin was enthusiastic about its style.

So he picked out a few passages and summarized them briefly. Without looking at the original, he tried to write the passages from memory, using his own words if he could not think of anything [13]. He then compared his essays with the original and found that neither his vocabulary nor his choice of words was at the desired level. In order to increase his vocabulary and reduce the spread of his verbal precision, he went beyond mere comparison and correction: he rewrote all the passages in verse [13].

It is fascinating that Franklin unfolded prose in verse to recognize the elements of beautiful style: It is like sending a beam of light through a dispersive prism to split it and see the beautiful colored light spectrum. He then tightened verses back into the unbound language to check whether he actually mastered these elements.

This is exactly what we can learn from him: We will use the pattern solutions to make the elements of physical thinking and the gaps in understanding visible, to fill them with understanding.

8.2 You Can Do that Too

The following “learning algorithm” represents an application of Benjamin Franklin’s idea to the exercises:

1. Find a sample solution.
2. Understand the solution.
3. Write it from memory. Compare it with the sample solution. Find all deviations.
4. Analyze and correct the deviations.
5. Repeat 3 and 4 until all deviations disappear.

The second step is to really understand the solution—only then you move on to the next step. The fourth step also means that you should find out the reasons for your deviations: Why did you make this one mistake? What did you not understand in the lecture that you made this one mistake?

With an example, we make the algorithm work. Let us say you have the following task to solve:

A point mass m slips without friction into a loop with diameter D . From which height h must the mass start so that it does not leave the track at the highest point of the loop?

Let us start with the transcription!

1. Find a sample solution: Very easy. Here is one:

At the highest point of the loop track, the weight and the centrifugal force of the point mass must balance each other:

$$mg = \frac{mv^2}{r} \Rightarrow v^2 = gr$$

The kinetic energy at this point is

$$E_{\text{kin}} = \frac{1}{2}mv^2 = \frac{1}{2}mgr$$

The total energy at the highest point of the loop path is the sum of the kinetic and potential energy:

$$E = E_{\text{kin}} + E_{\text{pot}} = \frac{1}{2}mgr + mgD$$

Because of the law of conservation of energy, this energy must be equal to the initial potential energy:

$$\begin{aligned}\frac{1}{2}mgr + mgD &= mgh \\ \frac{1}{2}mg\frac{D}{2} + mgD &= mgh \\ \frac{5}{4}D &= h\end{aligned}$$

2. Understand the solution: We are talking about the balance of forces. Apparently, at the highest point of the loop track, the downward force must balance the upward force so that the point mass does not fall.

Then one looks at the individual energies so that their sum can be equated with the potential energy at the level h .

Basically, only two things are necessary: energy conservation and balance of forces.

3. Write it from memory. Compare it with the sample solution. Find all deviations: I recall two steps were necessary: conservation of energy and balance of forces. At the highest point of the loop path, the forces were: the centripetal force pointing toward the center must balance the weight:

$$-\frac{mv^2}{r} = mg \Rightarrow v^2 = -gr$$

Then I will look at the kinetic energy:

$$E_{\text{kin}} = \frac{1}{2}mv^2 = -\frac{1}{2}mgr$$

The total energy at the highest point is therefore:

$$E = E_{\text{kin}} + E_{\text{pot}} = -\frac{1}{2}mgr + mgD$$

This total energy must balance the initial potential energy:

$$\begin{aligned}-\frac{1}{2}mgr + mgD &= mgh \\ -\frac{1}{2}mg\frac{D}{2} + mgD &= mgh \\ \frac{3}{4}D &= h\end{aligned}$$

4. Analyze and correct the deviations:

And here, poor fool, I stand once more, no wiser than I was before. Obviously the error is in the sign. What caused it? Apparently, centripetal force is not the same as centrifugal force. The reason for the error is the wrong understanding of what these forces are, not the sign itself. Consequently, I have to deal thoroughly with the rotating reference system, the inertial system and the inertial forces.

5. Repeat 3 and 4 until all deviations disappear

The difference between centripetal and centrifugal forces is now clear to me. Hopefully, I would not make the same mistake again....

8.3 The Art of Correct Copying

When editing the sample solution, you should break it down into the most basic units of understanding. The transitions between these units must be understood. Then you cover the sample solution and try to write it from memory. Even if you understand the solution 100%, you will often notice that you show some differences when comparing: These deviations are not necessarily errors, but can represent alternative calculations.

And this is exactly the strength of this method: **Like on a radar monitor, every deviation beeps and alerts you about your gaps in understanding**—metaphorically speaking, of course. This should make the weekly exercises less of a hassle for you and passing the exam should not be a problem. But there is a twist in this story in which you, as the protagonist, could suffer your at the lowest point: Exams do not consist entirely of the exercises you have already calculated. **Therefore, you should vary your problems.**

This means that you apply this method to thematically related tasks. If you have done several exercises with this algorithm, you can try to calculate the new unknown tasks on your own and skip steps 2 and 3, because you have already recognized the solution structure of such problems.

Although you have to solve a huge number of physical problems during your studies, it is all the more important that you also tackle different problems. In addition, something else is extremely important: **Try to exceed the sample solutions.**

Can it be simpler, faster or more elegant? In the first few semesters, you have not yet acquired the physics way of thinking and therefore you think more simply—this statement can of course be relativized. That is one of the reasons why

you spend so much time on your homework. But it also has the advantage that sometimes you get simpler solutions. Try to use this naivety to become better.

The origin of the word *art* can be traced back to the Pre-Socratics: they used it to refer to craftsmanship [23]. *The art of correct copying* should also be understood in this sense. The emphasis is on the craftsmanship approach to model solutions: All steps have to be learned and perfected one after the other and the thinking has to be practiced so that later on a valuable piece of art is created—your own solution.

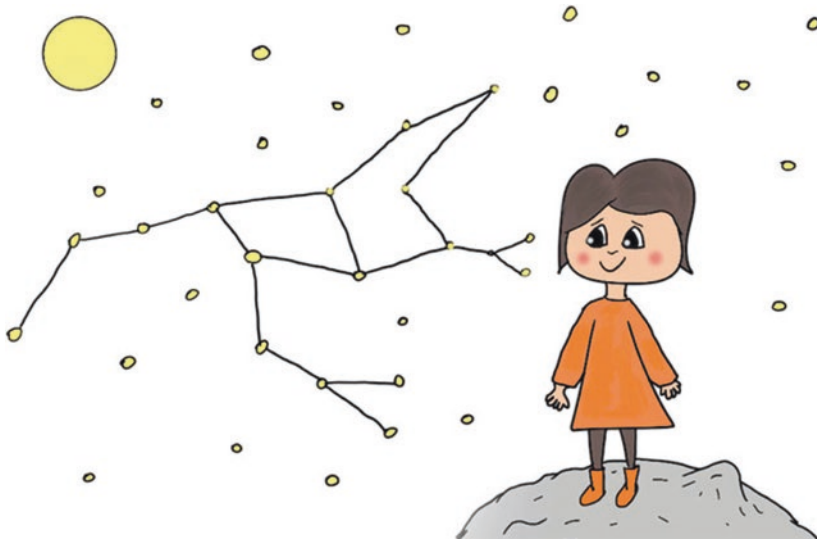
TODO

- Use the described learning algorithm.
 - Find a sample solution.
 - Understand the solution.
 - Write it from memory.
 - Analyze it and correct all deviations.
 - Repeat the process until all errors disappear.
- Vary your problems.
- Try to exceed the sample solutions.

How You Recognize Connections and Keep Them in Mind

9

Summary



Often you are in a local crisis: you do not understand the thinking behind the lecture. One does not know what the professor is aiming at or what the exercise problem is aimed at. Or you sit down to the exercises and do not find any approaches to the problems at all. But these local instabilities can easily adopt the dimensions of a global crisis. The result is typically self-doubt: *I know people*

who are much better than me and they have dropped out of physics. With the others everything looks so simple! Is it normal that I understand so little?

Local crises result from the inability to integrate the information received into what is already understood. You can recognize this inability during the lecture by the typical question of your seatmate: *Why are we doing this at all?*

In retrospect, this question can be answered because you see the big picture. In this chapter, you will therefore learn how to see it from the beginning and thus understand the connections faster.

9.1 Create a Map of Physics

You take a meaningful shortcut to the point where you have an overview of the entire landscape by creating a map of physics from the beginning. Whether it looks more like a written summary or more like a real map depends on your way of thinking. However, you should be able to locate the new knowledge on it. What does this mean? The following example should give you a feeling how to create your own map:

I clearly see three major fields of physics before me: (1) classical mechanics, (2) quantum mechanics and (3) relativistic mechanics.

1. Classical mechanics

In classical mechanics, one is mainly concerned with the movements of bodies described by Newton's laws: For example, if you throw an apple at an angle, you can calculate its trajectory exactly.

If you want to calculate the motion of an object that is in the same order of magnitude to the sun as the apple is to the gymnastics ball—the planet Jupiter, for example—you need Kepler's laws. For they describe the motion of a body around its central body: Like that of Jupiter around the sun.

History of physics is inserted in the apple anecdote: Newton sees an apple fall from a tree and wonders why the moon does not fall to earth. What follows is Newton's law of gravity, which combines the first mathematical descriptions of Kepler's laws, the motion of the moon around the earth and the earth's gravity.

With the law of gravity and optics—the science of light propagation that made telescopes possible—one could get a glimpse of the great mysterious universe: Cosmology and astrophysics developed.

Light propagates as an electromagnetic wave. Electrodynamics is concerned with electromagnetism.

Fundamental are also the main laws of thermodynamics, which—recognizable by their name—are dealt with in thermodynamics. The most important terms here are temperature, heat, energy, and entropy, which are used to explain heat machines, for example.

Classical mechanics explains the macro world. To describe the behavior of objects in the micro world, one needs quantum mechanics.

2. **Quantum mechanics**

If one takes the tools of quantum mechanics and works with them on thermodynamics, one enters the field of statistical physics.

From here you have a wide view of chaos theory. Quantum mechanics includes atomic, nuclear, particle physics, and many other disciplines that study phenomena at the quantum level.

3. **Relativistic mechanics**

This refers to the special and general theory of relativity. The special theory of relativity is the relativistic extension of Newton's laws: Here mass can change and velocities are close to the speed of light.

Important results of the special theory must be known: The speed of light is constant and $E = mc^2$.

The general theory of relativity explains gravity as a property of four-dimensional space-time.

4. **String theory or loop quantum gravity?**

Quantum mechanics plus special relativity yields quantum field theory.

Quantum physics plus general relativity results in a question mark that one would like to replace with string theory and loop quantum gravity.

Other subfields of physics, such as biophysics or computational physics, are located at the interfaces between physics and other sciences.

To outline your own map, you should look into the books in Chapter 12 section *Maps of Physics*.

9.2 Connect the Points on the Map

The creation of the map is extremely important: It provides you with a template with the points that you will combine in your studies lecture by lecture, task by task and arithmetic step by arithmetic step. And the question *why are we doing this at all?* disappears by itself.

But how exactly do you connect the points?

In Chapter 7, we have learned how to form a mental model. Every time you have formed a new mental model—that is, you have really understood a principle—you should ask yourself where it should go on the map: Is this an example of Newton’s second law? Or an application of the laws of energy and conservation? Locate the new knowledge so that you can navigate safely in your physics studies.

You should also combine it with your previous knowledge to understand the connection.

9.3 Connect the New Knowledge Constantly with Your Previous Knowledge

You are not a blank slate: the school certainly left several bruises and signed them in red. But this also means that you already have certain previous knowledge. Now use the power of your prior knowledge by building the new knowledge on it, because it can speed up the learning of new concepts [35].

Fortunately, you often combine what you have already learned with new concepts in physics. For example, the harmonic oscillator faithfully accompanies you until the end of your degree: Starting with the ideal spring, one later looks at the physical pendulum, coupled oscillations, movement near a potential minimum, Fourier analysis, lattice oscillations in solids, the quantum mechanical harmonic oscillator, polymer chains, quantum fields—and the list goes on.

The emphasis is on the word *constantly*, because now and then it is not enough: you have to keep connecting the new knowledge with your previous knowledge so that it is firmly anchored in your memory.

In spite of all efforts, it can and will certainly happen that something is not explained in greater depth and you simply have to accept some steps. What do you do then? Just trust that the points on the map can be connected afterward—but remain curious and persistent.

TODO

- Create a map of physics.
- Connect the points on the map.
- Keep connecting the new knowledge constantly with your previous knowledge.

How to Prepare for Exams Effectively

10

Summary



One of the reasons why the exams seem to be particularly difficult in the first semesters is the fact that at the beginning, you do not know how to prepare for

the exams at the university. Only later do you remember the first exams and say to yourself with a touch of pleasant remorse: If I had known that earlier!

Well, this chapter gives you that experience.

10.1 Written Examinations: PEAK—Premortem, Elimination, Automation, Exam (Premortem, Elimination, Automation, Klausur)

10.1.1 Premortem

One week after you have written your first exam, you will receive an email from your professor asking you to come to his office urgently. There you will learn that you have really flunked your exam: The worst exam he has seen in decades! His gaze pierces you and he says frostily: Please tell me how it happened.

This frightening scenario presents a useful form of exam preparation. The psychologist and Nobel laureate in economics Daniel Kahneman calls the presentation and analysis of such scenarios *premortem* and recommends to practice it especially before making important decisions [19].

Translated from Latin, *premortem* means before death: During a premortem analysis, you therefore imagine everything that can go wrong both during preparation for the exam and during the exam: Your roommate is studying music and has been on your nerves half the night. You understood everything, but your anxiety blocked the access to the cognitive machinery. Or there were too many problems in too short a time, so that your hand cramped up in the middle of the speed-writing exam.

There are many examples of what you can do wrong. During my studies, I asked my fellow students about their most common mistakes during exam preparation. Here are the results:

1. Started too late with the preparation.
2. Problems of the same type solved. This includes: Only easy tasks calculated so that they were overwhelmed by the difficulty of the exam.
3. Rely on the formula sheet if it was allowed in the exam.
4. Postpone: Little work done during the semester, so that you learned shortly before the exam to the point of exhaustion.
5. Spent too much time summarizing.

6. Focusing on one topic and ignoring the rest. This includes: Concentrating only on the problems and leaving out the theory.
7. Unusual but possible: Too much done, so that one has lost the overview.

So imagine as many worst-case scenarios as possible and transfer them to a checklist. Then we move on to the next step: elimination.

10.1.2 Elimination

Elimination in this context means that you detect exam-relevant problems and sort out the rest. This has nothing to do with laziness, it is a sober consideration of an optimization problem under the secondary condition of time shortage. What we want to optimize is the effective preparation for the exam. And here now follows the procedure:

The very first practical step is the procurement old exams and the corresponding solutions: The student committee, the Internet and students from previous years who have already written their exams with your professor are the first points of contact. As for the solutions, you should google them first of all. Often, however, not all solutions can be found on the Internet. Next, you should take a closer look at the textbooks: For example, the Halliday not only has problems at the end of each chapter, but there is even an additional solution volume (see chapter *Useful Resources* for more information).

However, it is possible that neither the Internet nor textbooks provide all solutions. In this case, you should pass on the problems to very smart fellow students: *Hey [name], it seems that this task is very important for our exam because [reason]. But I cannot get any further. I think you can solve it. Could you send me your solution then?* It is important that you mention two things: First, the reason why the problem is important because it speaks to the recipient's self-interest. Second, the expectation that the recipient can do something you cannot—solve the problem. Studies have shown that expectations can be used effectively to motivate people to act—both positively and negatively [5, 25].

But what if you cannot find any old exams? In Chap. 4, Sect. 4.2, we have mastered the art of research: Now use this skill to find exam-relevant problems. It also applies if you have found old exams—because additional exam-relevant problems do not hurt.

In the last step, you eliminate unimportant tasks. Suppose you are preparing for an exam in experimental physics. Normally, you have 12 series of exercises with 4–8 problems per series, so in total you have at least 48 exercise

problems. But only about a quarter of these problems are relevant for the exam, because—depending on your university—it will take 2–4 h. It is interesting to flirt with the idea that you only need to be able to calculate 12 problems to pass the exam. But the question arises: Which ones?

First of all, you sort out very easy problems, because they are far too trivial for an exam. Then you sort out very difficult problems, because—who would have thought—they are too complicated and arithmetically too elaborate to be calculated in the exam. The rest you judge with the following questions: Does this problem illustrate a physically meaningful concept? Is this a standard problem that every physicist must have calculated once? What are the preferences of the person who will design the exam? Maybe he/she has a weakness for oscillators because they vibrate so beautifully? Or maybe you can identify a tendency by looking at the old exams?

After the elimination comes the real work: automation.

10.1.3 Automation

The exam can be difficult for two reasons: On the one hand, because of the difficulty of the problems. On the other hand, because you do not have enough time to solve them, although you could solve all the exam problems. To avoid the latter, you should automate the process of solving them.

But before I describe it in more detail, I would like to warn you: **You should not prepare for exams in working groups.** Solving exercises together is okay. But you will write the exam on your own. Accordingly, you should prepare for the exam on your own. Having said that, we now come to the actual process of automation.

The starting materials for our preparation are the exercises, the old exams, the exam-relevant problems and the corresponding solutions, which we have found with the help of targeted research. **The first step is simply to solve these problems:** However, if you have been working on a problem for too long, do not hesitate to look at the sample solution. In the best case, you should already have the mental models and mathematical tools described in Chap. 7 at hand. However, if it was too much work for you to write them down for each problem during the semester, then you should do this now: Leave enough space in the margin to write down important formulas and the mathematical tools you need.

When you have solved all the problems, write down all the formulas and tools on a separate sheet of paper: So you create your own formulary. It is important that you create it yourself and do not use a ready-made formulary—even if it was

created by one of your fellow students. Because the actual sheet is completely irrelevant: It is solely about the process of cognitive reduction and abstraction of a task to a formula, a principle or a law.

The second step repeats the first one, sorting out all the problems you can solve independently, quickly and without looking at the sample solutions. In the next step, you calculate the remaining problematic **problems** and sort out those that you can successfully calculate. And so on until you can solve all **problems** independently and quickly: There must not be a single problem that you cannot solve.

The third step requires the concentration of a Jedi Master: You look at the conceptual formulation of each problem and go through the solution mentally. Because it is not enough that you have understood the tasks and can solve them: In the exam, you must solve them and related problems quickly. Therefore, you must not waste your time searching for solution ideas or suitable formulas in your head. Rather, you should be able to navigate safely in your mental formula world and know the quickest way to get from problem point A to solution structure B. And it is exactly this navigation that you stabilize, if you go through the solutions several times in your head. The mental models could then be the intermediate stations on your mental map: For example, to get to the solution of the problem of the ballistic pendulum, you have to go via energy and momentum conservation.

In the last step, you simulate the exam: Take a stopwatch (tap on the stopwatch symbol on your smartphone), set the duration of your exam and solve an old exam. You should expose yourself to mild stress [29]: Create conditions where you can hardly concentrate. For example, you could put on headphones and turn on the music you cannot stand.

Ideally, this should take place 2 days before the exam, so that you can spend a day relaxing and staring at your solutions in a hypnotic state.

10.1.4 Exam

Too much anxiety shortly before the exam can lead to a mental block [38]: If you do not let go, then all the hard work during the semester and the preparation is useless. On the other hand, it can also be a good thing: The blood starts to pump faster and the brain is better supplied with blood—you are ready to go to the top of your cognitive performance.

To prevent the anxiety from leading to a mental block, you should do something that captures your emotional and mental distractions like a converging lens

and focuses them on a point in the here and now: For example, listen to one of your favorite songs, play on your smartphone or try pen spinning—rotate a pen around your thumb. **So find an activity that requires your concentration in a pleasant or playful way.** It is also a good idea to write down your feelings and thoughts 5 or 10 min before the exam starts [9].

Now follows the actual procedure of **problem** solving. Should one start with easier or more difficult problem? **For the following three reasons, you should start with the easier tasks:**

First, there is the risk that you will spend too much time on a difficult exercise—regardless of whether you will successfully complete it or not. Second, you may not be able to solve it because it is difficult. If this happens, you will panic, rush and therefore make attentional errors. Lately, easier exercises are easier to solve—who would have thought it! This gives you a feeling of achievement. It makes you more confident about the rest. But what happens when you are completely lost in front of the exam sheet?

Then you should raise your hand and ask a question: Whether or not you actually have a question. Because it is much more about the process of asking: Either the supervisor could give you a hint or you could be stimulated in the right direction of thinking by this process.

TODO

- Imagine as many worst-case scenarios as possible and transfer them to a checklist.
- Procure old exams, other exam-related tasks and the corresponding solutions.
- Eliminate unimportant exercises.
- Automate the process of problem solving.
 - Solve all problems as usual.
 - Repeatedly solve and sort the tasks that you already know how to do.
 - Go through all the solutions in your head.

Just before and during the exam:

- Do something that challenges your concentration in a pleasant or playful way.
- Solve the easier problems first.
- Raise your hand if you are clueless

10.2 Oral Examinations

The oral examination is a completely different examination situation than the written exam, because it usually takes 15 to 45 min. Therefore, no difficult arithmetical problems or even lengthy derivations are asked here, but rather comprehension questions and short calculations.

Accordingly, your starting point for the preparation looks different: You only need the exercises and their solutions and a very good transcript of the lecture in the sense of Sect. 6.1.

10.2.1 Exercises

Here you should know how you would solve difficult problems. And you should be able to solve simple tasks easily. The difficult tasks are all about which ideas or steps would be necessary to solve them and not about the explicit calculation.

For example, a question could be as follows: Can you derive X ? Where X stands for any law. Then you sketch the ideas of a derivation: Under which boundary conditions do you consider the problem? With which quantities do you start? What do you use in the derivation?

10.2.2 Preparation

All the raw material for the preparation consists of the following things:

1. A very good transcript.
2. A list of questions: Ideally, you should already have a list of questions collected throughout the semester, as I describe them in Sect. 6.1 *Encryption is secure*. However, if you have not yet created it, you should edit your transcript thoroughly and write down possible exam questions.
3. The noted solution ideas to the difficult tasks and solutions of the simple arithmetic bites.
4. A “personality profile” of the examiner: in what field is he/she specialized? Does he/she have a preference for certain physical concepts? Is he/she a “conservative” physicist who wants to hear exact definitions and justifications, or is he/she more the type who values something like intuition behind physics?

The first thing to do is to edit this source material very thoroughly: understand everything in your notes completely. Answer each question in the list. Master all simple exercises and remember the rough solution of the difficult ones. The personality profile could give you clues as to what might be particularly important.

Second, you reduce it to the most important ideas, formulas, sentences, etc.

Third, you create a local map in the sense of Chap. 9, which not only summarizes the reduced material, but also presents it in a clear way. The landscape of this map can be individually adapted—but it must serve a purpose: You should be able to recognize connections and remember them quickly. Imagine it like this: You are flying a helicopter and you are looking down from above at what you have learned. If you are asked for a context in the exam, you should know exactly which course you need to take mentally in order to quickly locate it on the map and fly there. Figure 10.1 shows an example of a part of one of my maps.

Next you take—as in the preparation for the written exam—a stopwatch and a few questions from your list and simulate some test flights. Again, you should speak out loud to yourself to verbally show what you can do—and you can only do what you repeat over and over again.

Finally, if you have the opportunity, you should choose the date of the exam so that you are one of the first to be tested. Why is this practical aspect important? Because the examiners are probably not tired yet – it is simple as that. Because fatigue and other factors such as hunger have a great influence on the assessment process [19].

10.2.3 Overcoming Excitement

The anxiety before and during the oral exam is different from that in the written exam, because you are asked directly by the examiner and must be able to express your thoughts clearly. Therefore, you should also keep your anxiety at bay differently than in the exam.

But here too, a slightly modified version of premortem analysis helps: **Imagine the worst oral exam you could ever take.** You are late. The examiners are annoyed. Every sentence you say is immediately criticized. At the end of the exam, you can read it unmistakably in the dissatisfied expression of the examiner: failed.

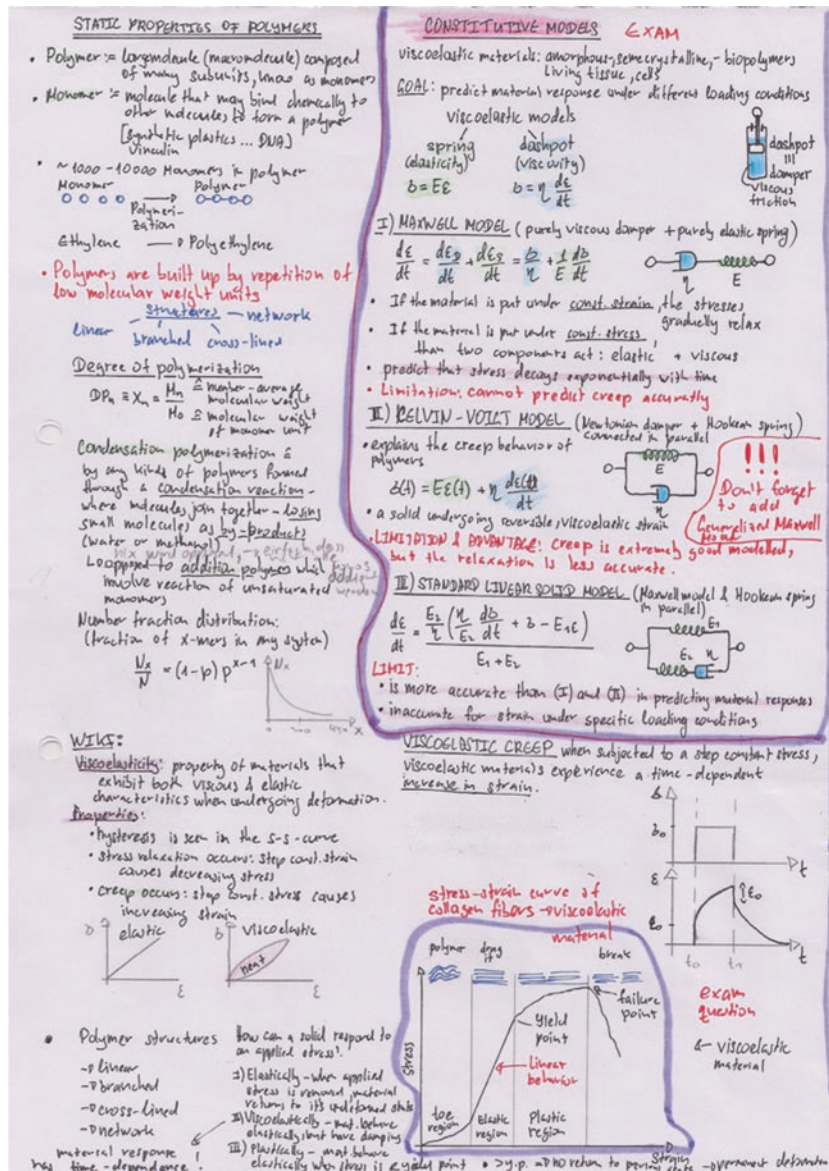


Fig. 10.1 A part of my map for the lecture Soft Matter Physics

Now imagine what you could do about it. How do you act? Do you maintain control and answer calmly and accurately? Or do you take the offensive, but at the same time show your competence?

Imagine this scenario as real as possible. Because then you are prepared for the worst, which probably would not happen anyway, because the professors are usually nice and helpful during the exams—and if not, you know what to do.

Another possible mental tool for coping with the anxiety I call ego dissolution. Both in the preparation and in the oral examination, there is a danger of being far too focused on yourself: In the preparation phase, it can manifest itself in imagining very unproductive horror scenarios and in paralyzing exam nerves. During the exam, it can even have a verbal paralyzing effect so that you cannot get a word out of you.

In practical terms, ego dissolution means that you shift your concentration to the examiners. Instead of seeing yourself fully lit up in your thoughts on stage, you turn the spotlights on the examiners: For example, formulate *What will they think of me?* in *How can I express my answers so precisely and clearly that they do not get bored?*

10.3 After the Exam

Immediately after the exam, you feel freed from the heavy chains of exam preparation. At this moment, you do not want to hear anything more about the exam and only want to enjoy your freedom. Here is why you should not do it yet.

The time immediately after the exam is the best time to reflect on it and to write down your mistakes so that you do not make the same mistakes in the next exams. Take a writing instrument and a writing pad and analyze what you have learned. The following questions can help you:

- What questions, problems and topics did I expect and what did I actually get?
- What was I surprised about? Why was I surprised?
- What did I find particularly difficult? Why?
- What mistakes did I make? Why?
- What do I learn from this exam?

Learning does not end with the grading: it is only a stopover on the way to competence, where you should reflect on your mistakes and experiences and enter them into the timetable. But the train has not left the station if you have really

blown your exam, because often you learn best when you have done something wrong: pick yourself up, understand your mistakes and do it better.

Learning for the exam and for yourself is not very useful—the latter sounds vague and besides you never learn for yourself: from being tested to the fact that you communicate with others. Rather, it makes more sense to work on your competence: But this is impossible if you only study for good grades. And that brings us to the last chapter.

TODO

- Edit your source material thoroughly.
- Reduce it to the most important ideas, formulas, sentences, etc.
- Create a clear map.
- Simulate the oral exam.
- Choose the time of the exam wisely.
- Apply the premortem analysis here as well.
- Dissolve your ego.
- Reflect in writing as soon as possible after the examination.



Sustainable Competence Instead of Short-Term Competition

11

Summary

How does one become sustainably competent? What does that mean anyway?

In the course of the book, I have scattered the idea of craftsmanship: Whether you want to solve physical problems, overcome exam nerves or even copy the sample solutions: Do it in such a way that you recognize frequent mistakes and master every step to finally produce a high-quality piece of work—your competence.

This also means that when you study physics, you do not orientate yourself on the greed for good grades or on comparing yourself with your fellow students, but navigate with the help of this idea of competence, because this can also help you in the difficult phases of your studies – but not the striving to collect good grades like trophies: once you get a bad grade and you are psychologically at the end of your tether. It is different if you learn sustainably.

Then you know that you could and should learn from every failure, because you will become better, more competent. You know that you should work on something that you still lack or that needs to be overcome: Like the fear of oral exams, for example, which is especially frustrating when you actually know exactly what you are talking about. For this case, as well as for the whole study in general, I would like to give you one last advice:

Face the problem—step by step, continuously. In psychology, it is called confrontation therapy [16]. Stage fright? Give a presentation first to a dog, then to a few friends and then to some strangers. Afraid that a bachelor's thesis in theoretical physics is too difficult for you? Then write a little term paper in it first and see how it goes on: But do not let yourself be intimidated right at the beginning of your project.

As it may be that in another universe, a copy of you has just made a chain of very unfortunate decisions and has never faced its problems, never learned from its mistakes and therefore never experienced how exciting studying physics can be.



Useful Resources

12

Summary

In this chapter, you will find very useful books, online lectures and courses. These recommendations are highly subjective but tested by the many hours of reading and suffering.

Mathematics (analysis, linear algebra, etc.)

- www.onlinetutorium.com

The Analysis I part is particularly recommended. If I could travel into the past and study physics from scratch, I would start with this tutorial from the beginning. Watch all videos in the Analysis I part in the first semester: They give you a solid foundation for your first math exam. Also, Chapter 2 *Complete Induction* helped me a lot.

- **Mathematics (Mathematik) by T. Arens, F. Hettlich, Ch. Karpfinger et al.**

Here, Part III *Linear Algebra* is particularly recommended. At some point in time, you will need to calculate the volume and surface area of an n -dimensional sphere for the lecture in statistical physics—you will find this on page 1179 *Volume and surface area of an n -dimensional sphere*.

- **Mathematics for engineers and scientists (Mathematik für Ingenieure und Naturwissenschaftler) by L. Papula**

These volumes are for anyone who wants to learn and master the mathematical craft. The theory is clearly explained and presented with a vast number of practical examples and tasks. Especially the chapter *Ordinary differential equations* in Volume 2 is recommended.

- **Analysis by Otto Forster**

Perfect for the exercises, because there are separate exercise books with solutions: Learn and practice.

- ***The yellow arithmetic book (Das gelbe Rechenbuch)* by Peter Furlan**

The yellow arithmetic books are also particularly suitable for the exercises, because the three volumes calculate everything in single steps. In addition, many examples are calculated in detail, which professors like to present as exercises. You should have a look at Section 2.7 *Sequences (Folgen)* in Volume 1 if you are going to deal with sequences.

- ***Calculus* by S. L. Salas and Ainer Hille**

Especially Chapter 16 *Gradients, Extreme values, Differentials* and Chapter 18 *Curves and surface integrals* stand out didactically. The subchapter 16.10 *The reconstruction of a function from its gradient* helped me a lot in the second semester.

- ***Ordinary differential equations (Gewöhnliche Differentialgleichungen)* by Harro Heuser**

If you not only want to solve ordinary differential equations, but really understand them, then this is the book for you. At the end of each chapter, you will find the wonderful and highly interesting *Applications* section, which describes many classical physical models such as the Brachistochrone or Kepler's equation of planetary orbit. But just because of Heuser's entertaining *Historical Notes*, you should read it.

- ***Analysis 2* by Stefan Hildebrandt**

- Hildebrandt is particularly helpful for one topic: manifolds. Grab the book and open Chapter 4 *Equation Defined Manifolds* when it is your turn.

Maps of physics

- ***The Character of Physical Law* von Richard P. Feynman**
- ***Key concepts in physics: From the Newton axioms to Hawking radiation (Schlüsselkonzepte zur Physik: Von den Newton-Axiomen bis zur Hawking-Strahlung)* by Klaus Lichtenegger**
- ***The Theoretical Minimum: What You Need to Know to Start Doing Physics* by Leonard Susskind**

Experimental physics

General textbooks

- **Halliday Physics**

The book is ideal for solving the problems. The theory is clearly explained, so that one begins to develop the physicist way of thinking.

- ***Physics: Textbook and exercise book (Physik: Lehr- und Übungsbuch)* by Douglas C. Giancoli**

Giancoli is Halliday's twin brother: this suggests that everything that can be said about Halliday also applies to Giancoli. So choose one of the two books from the beginning and work through it consistently throughout the semester.

- **Textbooks of experimental physics by L. Bergmann and C. Schäfer**

I have a great respect for Bergmann and Schäfer: the books are old school. They are detailed, solid and inspiring at the same time.

- ***Schaum's Theory and Problems of Modern Physics***

Exam preparation? Then grab the book!

- **Standard textbooks by Wolfgang Demtröder**

Actually, they are excellent textbooks. But they are only excellent if you are already somewhat advanced.

Solid state physics

- ***Solid state physics (Festkörperphysik)* by R. Gross and Achim Marx**

- ***Solid State Physics* by N. W. Ashcroft and N. D. Mermin**

- ***Solid state physics* by C. Kittel**

All three works are solid standard textbooks: they are understandable, vivid, and comprehensive. So you are not doing anything wrong when you read one of the books. (Although I prefer Gross and Marx, because it was more interesting to read.)

Theoretical physics

General textbooks

- **Textbooks by Wolfgang Nolting**

Nolting can save your life: No matter if you are reworking your lecture or despairing of the tasks—it is the book par excellence. The theory is explained in an extremely understandable and detailed way. In addition, at the end of each chapter, you will find didactically perfectly selected problems that test and deepen your understanding.

- **Textbooks by Torsten Fließbach**

These textbooks are understandable and concise: if you do not want to spend your time with digressions and excursions, then this is the best choice.

- **Landau and Lifschitz**

Landau and Lifschitz is not for beginners. But the presentation of the physics is so elegant and masterly that I cannot leave it unmentioned.

Quantum mechanics

- ***Quantum mechanics (Quantenmechanik)* by D. J. Griffith**

Quantum mechanics is so abstruse that the Copenhagen interpretation can be summarized in one sentence: *Shut up and calculate!* [24] That is why one should master the mathematical tools of quantum mechanics. Griffith will teach you exactly that.

Statistical physics

- ***A Modern Course in Statistical Physics* by L. E. Reichl**

Since I only discovered the book at the end of my studies, but did not read it completely, I can only judge the following two chapters: *Equilibrium Statistical Mechanics II* and *Brownian Motion and Fluctuation and Dissipation*. Especially the ideal Bose-Einstein gas, the ideal Fermi-Dirac gas and the fluctuation-dissipation theorem are explained in detail.

Particle physics

- ***Introduction to Elementary Particles* von David Griffiths**

During one of my highly unscientific experiments, I took a course in particle physics to test the application of the Pareto principle to physics. Result: Grade 3.3, so maybe I should not recommend books on particle physics. But I think the book is so good that I cannot help it.

Inspiring and motivating books

I did not mention the famous and excellent *The Feynman Lectures On Physics* above because I found it hard to read at first. Nevertheless you should take a look at this fantastic work.

It happens during studies that you can get lost in countless calculations, derivations and your specialization and forget the reason why you actually study physics. To get back to the roots of thinking about physics and to cheer up in the more difficult phases of studying, I keep returning to the following books:

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- *Surely You're Joking, Mr. Feynman* by **Richard P. Feynman**
 - *The Last Man Who Knew Everything* von **Andrew Robinson**
 - *Heisenberg Probably Slept Here* by **Richard P. Brennan**

But of course there are many more inspiring books and all books by Richard Feynman are recommended.

What You Learned From This *Essential*

- You study together, not against each other.
- Studying to be competent in the long term helps in the difficult phases of studies.
- You have to trust that at some point everything will make sense.

References

1. Bevelin, P.: Seeking Wisdom: From Darwin to Munger. PCA Publications LLC, Malmö (2007)
2. Bloch, R.L.: My Warren Buffett Bible. A Short and Simple Guide to Rational Investing: 284 Quotes from the World's Most Successful Investor. Skyhorse Publishing, New York (2015)
3. Brandt, S., Dahmen, H.D.: Mechanik: Vom Massenpunkt zum starren Körper. Springer, Wiesbaden (2016)
4. Buzan, T.: Speed Reading. David & Charles, Exeter (1988)
5. Cialdini, R.B.: Influence: The Psychology of Persuasion. Collins, New York (2007)
6. Cirillo, F.: The Pomodoro Technique. Simon and Schuster, New York (2014)
7. Cognition, T., Laboratory, A. N.: What are the effects of handwriting on cognitive development? <http://indiana.edu/~canlab/handwriting.html>
8. Currey, M.: Daily Rituals: How Artists Work. Knopf, New York (2013)
9. Doherty, J.H., Wenderoth, M.P.: Implementing an expressive writing intervention for test anxiety in a large college course. J. Microbiol. Bio. Educ. **18**(2) (2017)
10. Dunlosky, J., Rawson, K.A., Marsh, E.J., Nathan, M.J., Willingham, D.T.: What works, what doesn't. Sci. Am. Mind **24**(4), 46–53 (2013)
11. Ericsson, K.A.: The Road to Excellence: The Acquisition of Expert Performance in the Arts and Sciences, Sports, and Games. Psychology Press, New York (2014)
12. Ferriss, T.: Tools of titans: The Tactics, Routines, and Habits of Billionaires, Icons, and World-Class Performers. Houghton Mifflin Harcourt, Boston (2016)
13. Franklin, B.: Benjamin Franklins Leben. <http://gutenberg.spiegel.de/buch/benjamin-franklins-leben-5789>. Zugegriffen: 24. Apr. 2018
14. Gleick, J.: Genius: The Life and Science of Richard Feynman. Vintage, Santa Monica (1992)
15. Heublein, U., Richter, J., Schmelzer, R., Sommer, D.: Die Entwicklung der Schwund- und Studienabbruchquoten an den deutschen Hochschulen. In: HIS: Forum Hochschule Bd. 3. (2012)
16. Hofmann, S.G.: Einführung in die moderne Kognitive Verhaltenstherapie: Psychotherapeutische Lösungsansätze. Springer, Wiesbaden (2013)
17. Hofstadter, D.R.: Gödel, escher, bach. Vintage Books, New York (1980)

18. Isaacson, W.: Benjamin Franklin: An American Life. Simon and Schuster, New York (2003)
19. Kahneman, D.: Thinking. Fast and Slow. Macmillan, New York (2011)
20. Kieffer, G., Kaiser, H.J., Hilke, A.: Einführung in die Psychologie. Gehlen, New York (1980)
21. Kurnaz, M.A., Eksi, C.: An analysis of high school students' mental models of solid friction in physics. *Educ. Sci. Theory Pract.* **15**(3), 787–795 (2015)
22. Langer, E.J., Blank, A., Chanowitz, B.: The mindlessness of ostensibly thoughtful action: The role of placebic information in interpersonal interaction. *J. Pers. Soc. Psychol.* **36**(6), 635 (1978)
23. Löbl, R.: *Techné: Untersuchung zur Bedeutung dieses Wortes in der Zeit von Homer bis Aristoteles*. Königshausen & Neumann, Würzburg (1997)
24. Mermin, N.D.: Could Feynman have said this. *Phys. Today* **57**(5), 10 (2004)
25. Miller, R.L., Brickman, P., Bolen, D.: Attribution versus persuasion as a means for modifying behavior. *J. Pers. Soc. Psychol.* **31**(3), 430 (1975)
26. Mueller, P.A., Oppenheimer, D.M.: The pen is mightier than the keyboard: advantages of longhand over laptop note taking. *Psychol. Sci.* **25**(6), 1159–1168 (2014)
27. Newport, C.: *Deep Work: Rules for Focused Success in a Distracted World*. Hachette UK, New York (2016)
28. Ng, A.: Deep Learning Specialization. Coursera. <https://www.coursera.org/learn/deep-neural-network/lecture/DHNcc/hyperparameters-tuning-in-practice-pandas-vs-caviar> . Zugriffen: 15. Apr. 2018
29. Oakley, B.A.: *A Mind for Numbers: How to Excel at Math and Science (even if you flunked algebra)*. TarcherPerigee, New York (2014)
30. Pavlov, I.: *Lectures on Conditioned Reflexes*, vol. 1 & 2. International Publishers, New York (1928)
31. Schwartz, D.L., Martin, T.: Inventing to prepare for future learning: The hidden efficiency of encouraging original student production in statistics instruction. *Cogn. Instr.* **22**(2), 129–184 (2004)
32. Schwartz, D.L., Chase, C.C., Oppezzo, M.A., Chin, D.B.: Practicing versus inventing with contrasting cases: The effects of telling first on learning and transfer. *J. Educ. Psychol.* **103**(4), 759 (2011)
33. Stigler, S.M.: Regression towards the mean, historically considered. *Stat. Methods Med. Res.* **6**(2), 103–114 (1997)
34. Wammes, J.D., Meade, M.E., Fernandes, M.A.: The drawing effect: Evidence for reliable and robust memory benefits in free recall. *Q. J. Exp. Psychol.* **69**(9), 1752–1776 (2016)
35. Warren, S. L.: Make It Stick: The science of successful learning. In: *Education Review//Reseñas Educativas* 23 (2016)
36. Wawrinowski, U.: *Grundkurs Psychologie: eine Einführung für Berufe im Gesundheitswesen*. Bardtenschlager, München (1985)
37. Wittgenstein, L.: *Tractatus Logico-Philosophicus*. Routledge, London (2013)
38. Zeidner, M.: *Test Anxiety: The State of the Art*. Springer Science & Business Media, New York (1998)